User Inputs

Accelerator Design | Design Constraints

Arithmetic Unit Library

Add8 | Sub10 | Mul11 | ...

Design Space Construction → **Initial Sampling** → **Synthesis & Execution** → Initial Dataset $\mathcal{D}_0$ → **Models Training PPA & Accuracy GNNs**

Initial models & measured references

Self-Evolving DSE

Search

Candidate Generation (NSGA-III)

Surrogate Prediction

Pareto Ranking

Candidate Selection

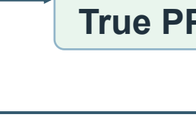
Enough Candidates?

No

Yes

Surrogate Models

Configuration Graph G



PPA GNN

Area / Power Readout

Path-Aware Readout (Latency)

$\hat{A}$

$\hat{P}$

$\hat{L}$

Accuracy GNN → Per-Image SSIM → Mean Accuracy

Local Calibration

Measured References

Latency Correction

$\tilde{L}(x) = L(r^*) + \hat{L}(x) - \hat{L}(r^*)$

Predicted Objectives $[\hat{A}, \hat{P}, \tilde{L}, \widehat{Acc}]$

Evaluation

Selected Configurations

Logic Synthesis True Area / Power / Latency

Application Execution True SSIM

Measured PPA & Accuracy y_t

Budget Reached?

No

Yes

Update

Surrogate Retraining | **Reference Update**

Updated models & references

Final Search Fixed Models

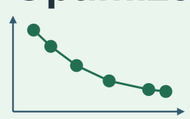
Pareto Front True PPA & Accuracy

Constraints Met?

No

Yes

Optimized Approximation



Parameter 1 | Parameter 2 | ...